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Reflection-equivariant representation

How to separate lateralized from symmetric gait by construction?

On source-video-disjoint folds, does building the signed left-minus-right axis to be antisymmetric BY CONSTRUCTION (an encoder whose readout the anatomical mirror is guaranteed to negate) separate lateralized gait (stroke, hemiplegic cerebral palsy, early Parkinson’s) from symmetric gait (myopathy) better than the standard `ea59fea0` encoder that was only allowed to learn that behavior, measured by item 05’s frozen signed-decodability and mirror-slope instrument at a pre-registered margin?

If you want to actually run this, see `METHODOLOGY.md`.

What actually happened (read this before anything else on this page)

Everything below the next two sections is the original PROPOSAL. The experiments have since been run, and their measured outcomes are recorded in `IMPLEMENTATION.md`, which is authoritative. Three experiments attacked one shared question and returned three verdicts that mean three DIFFERENT things. Never collapse them into “it did not work”.

Experiment	What it changed	Endpoint	Verdict
Idea 05, <code>nb_05a</code>	nothing; reads out of the frozen encoder	ridge R-squared on a signed target	INFORMATIVE NULL
Idea 9 Arm 1, <code>nb_09a</code>	the readout’s shape only; encoder still frozen	ridge R-squared on a signed target	ARTIFACT (side-agnostic nuisance control fired)
Idea 9 Arm 2, <code>new_nb_09_00..03</code>	the encoder itself, during the full curriculum	label-free mirror residual rho	NO CREDIT

Two scales before the numbers arrive. R-squared runs from 1.000, a perfect linear read, down through 0.000, no better than predicting the mean, and goes **NEGATIVE** when the fit is worse than that. rho is Arm 2’s label-free mirror residual, on a scale where **0 is mirror equivariant, the best value, and 4 is mirror blind, the worst value**.

Here is what each word means, because these three are routinely confused.

- **Informative null** (Idea 05) means the measurement was **VALID** and the answer was **NO**. On the authoritative checkpoint `ea59fea0`, the learned lane A scored R-squared -0.602, the untrained floor C scored -0.156, the side-blind pooled control D scored -0.131, the raw-coordinate ceiling B scored 1.000, and the measured anatomical-mirror slope was -0.741, outside the flip band. Two controls beat the treatment. A question was asked and answered.
- **Artifact** (Arm 1) means the measurement was **NOT ADMISSIBLE EVIDENCE** about sides at all, because the side-blind lane E (-0.066) outscored the antisymmetric treatment A prime (-0.206) by 0.140. The claim is

WITHDRAWN rather than answered, which is a WEAKER epistemic state than a null, not a stronger one. The wiring slope came back at exactly -1, so the head really was antisymmetric and this is not an implementation bug.

- **No credit** (Arm 2) means the effect is REAL and LARGE and improved 18 of 18 source videos, at about 7.1 times the control's seed spread, but a preregistered guardrail failed and its failure supplies a competing explanation, so the effect is not credited. "No credit" is not "no effect".

Two facts to carry away. First, each experiment closed one escape route from the previous one: Arm 1 removed "the readout was the wrong shape", and Arm 2 removed "the encoder was never asked", because the encoder demonstrably CAN be driven to ρ 0.059. Second, the binding constraint turned out to be the COHORT rather than the model or the readout: only 7.5 percent of the labelled target's variance lies between the 18 source videos, against a preregistered 30 percent, and that number is measured in **nb_09a** (Arm 1), not in Idea 05 and not in **nb_05b**. In all three experiments the informative element is a CONTROL rather than the treatment.

None of the three licenses a clinical claim, a statement about unseen videos, or any equating of ρ with performance. Every score is transductive.

Reified implementation (start here to run it)

This proposal is now reified as runnable code. The build log, the exact numbers each notebook writes, the hardened decision rule, and the register of SUPERSEDED claims live in IMPLEMENTATION.md. That file supersedes the proposal-level worked-example gates in the "The decisive experiment" section below.

CHECKPOINT NOTE. The standard frozen encoder on this page, in every method step, comparator, endpoint, gate, and figure caption, is the curriculum-final target encoder whose fingerprint prefix is **ea59fea0**. Every measured Idea 05 and Idea 9 Arm-1 number is bound to it. This proposal was originally drafted against an earlier **d0acc262** checkpoint, same configuration on a differently extracted pose cache, which is now SUPERSEDED; it is named here once, as history, so a stale number is recognizable on sight. Any document quoting $A = -0.187$, $C = +0.147$, $D = -0.014$, or a mirror slope of -0.343 as a current result is reading that superseded bundle and is wrong.

The idea is split into two arms, judged on Idea 05's frozen instrument. Four notebooks carry Arm 2's real run. All the long file and directory names are listed once here, so that the prose below can use short names:

notebooks

```
nb_09a_antisymmetric_readout_probe.ipynb
nb_09b_equivariant_retrain.ipynb
new_nb_09_00_methodology_and_contract.ipynb
new_nb_09_01_mechanism_and_smoke_validation.ipynb
new_nb_09_02_real_multiseed_equivariant_training.ipynb
new_nb_09_03_evaluation_results_discussion.ipynb
```

result bundles, under work/artifacts/<mode>/

```
idea9_antisymmetric_readout_result.json
idea9_equivariant_retrain_result.json
idea9_arm2/
```

- Arm 1 (zero-retrain readout): **nb_09a** applies the antisymmetric head to the EXISTING frozen **ea59fea0** encoder without retraining, and asks whether the antisymmetry constraint alone decodes the signed axis. Verdict: ARTIFACT (side-agnostic nuisance control fired). This notebook also measures the cohort fact that explains the whole family, the between-source variance fraction of 0.075 against a threshold of 0.30. It writes the Arm 1 readout result JSON listed above.
- Arm 2 (equivariance-coupled retrain): **nb_09b** adds a label-free equivariance penalty to the training loss and runs a D to D0 to E1 ablation ladder across seeds. This is the only arm that changes the encoder. It is the smoke-scale plumbing scaffold on a synthetic cohort, so its ladder numbers are PLUMBING ONLY and are not evidence about any checkpoint. Its two-part credit rule is also superseded. It writes the equivariant-retrain result JSON listed above.
- Arm 2's real run happened in a separate series, not by following **nb_09b**'s section-8 recipe, because that recipe's absolute equivariance term is satisfiable by shrinking the trainable head. The series is **new_nb_09_00** through **new_nb_09_03**, named in full above and run in that order, using a scale-invariant term and a parameter-free endpoint. Verdict: NO CREDIT. The endpoint is ρ , a label-free mirror residual whose scale runs from 0, meaning mirror equivariant and best, to 4, meaning mirror blind and worst. ρ fell from a control mean of 0.462 to 0.059 against a control seed spread of 0.057, with 18 of 18 source videos improving and the head's output

scale GROWING rather than shrinking, so the degenerate solution is excluded. But the `feature_std` guardrail regressed from 0.400 to 0.371 against a seed spread of 0.008, about 3.51 times the spread, and because a reflection-consistency term also removes variance, that regression is a competing explanation for the gain rather than an unrelated cost. Two protocol deviations are recorded: 3 seeds run against 5 registered, and the grouped five-class guardrail was **not evaluable** on this cohort (never say it failed). Its outputs land in the `idea9_arm2/` directory listed above.

- The `new_nb_09_01` mechanism numbers are SYNTHETIC FIXTURES (30 synthetic sequences, 30 epochs), not gait results. They calibrate the endpoint and expose the shrink-the-head failure mode. Fixture magnitudes sit near rho 3.9 whereas the real control lands near 0.462, and the notebook explicitly retracts any reading of fixture magnitudes as a forecast of real ones.
- Futures and reach: `nb_09c_futures_and_reach.ipynb` (no torch) rehearses the five possible verdicts through the same rule and scaffolds the non-clinical multi-view reach check. Result JSON: `idea9_futures_bundle.json`.

Each notebook is emitted by a committed builder script in this folder (`_build_nb_09a.py`, `_build_nb_09b.py`, `_build_nb_09c.py`). The hardened gates are: beat $\max(D, C)$ by 0.05 (the binding bar, with the untrained floor C as the real constraint), beat a capacity-matched control A_c by 0.05 (the same head with the symmetric left-plus-right path added, so a win is attributable to the antisymmetry constraint and not to head capacity), a permutation null on the head, an absolute nuisance-control gate $\text{abs}(E) < 0.05$ on a Lane E that is symmetrized over the anatomical mirror so it is genuinely side-blind, a y-quality gate, and the exact wiring-swap identity check. The floor C's own permutation null is still reported, but only as floor characterization (a strong floor just raises the binding bar); it is not a claim-withdrawing gate. See IMPLEMENTATION.md for the full rulebook.

The question in plain words

Think about how a healthy person walks. The left leg and the right leg do almost the same thing, just half a stride apart. Walking is nearly a mirror image of itself, left to right.

Some walking problems break that mirror on one side only. A stroke weakens one side of the body, because the nerve wires from the brain cross over to the opposite side (the crossover point is called the pyramidal decussation, Natali/Javed StatPearls, PMID 30571044, 30521239). Hemiplegic cerebral palsy comes from a one-sided injury to the developing brain's leg fibers, so one leg is stiff (Volpe 2009, PMID 19081519; Back 2007, PMID 17261726). Parkinson's disease usually starts on one side, because the brain loses its dopamine cells on one side first (Riederer and Sian-Hulsmann 2012, PMID 22367437). All three show up on a stick figure of the body as a left-versus-right difference. Myopathy is different: it is a muscle disease that weakens both sides about equally (Barohn 2014, PMID 25037080), and on a stick figure it shows NO real left-right difference compared to healthy people (Xiong 2023, PMID 37525241).

So there is one useful measuring stick that tells most of these conditions apart: how different the two sides are, and which side leans. Item 05 (a sister proposal in this folder) asked whether our existing trained model already carries a signed version of that measuring stick. "Signed" just means the number can be positive or negative, so it does not only say "how different" but also "which way," left or right.

This proposal takes the next step. Instead of hoping a normal model happened to learn the mirror behavior, we BUILD the measuring stick so that flipping the body left-to-right is guaranteed to flip the number's sign. That property has a name: reflection-equivariance. In plain words, reflection-equivariant means a mirrored input always makes a mirrored output, because the wiring of the model forces it, not because the training happened to teach it.

Here is the everyday picture. Imagine a scale that weighs "left minus right" leg activity. If you swap the person's left and right sides (a mirror), an honest left-minus-right scale must show the exact opposite reading: same size, opposite sign. We are going to hard-wire the scale so it can do nothing else. A standard model, by contrast, only "hopes" to behave this way if it saw enough examples.

Words you will need

- Skeleton: a moving stick figure traced over a walking person, made of body joints. It hides the face and clothes and keeps only how the body moves.
- Token: the smallest chunk the model reads. Here, one joint watched over a short slice of time.
- Embedding: a short list of numbers that acts like a fingerprint for a piece of input. Similar inputs get similar fingerprints.
- Encoder: the part of the model that turns raw joint positions into embeddings.
- JEPA: Joint-Embedding Predictive Architecture. A learning trick: hide part of the input and predict the hidden part as a fingerprint, not as exact pixels or coordinates.

- Signed axis: a single number that can be positive or negative. Sign tells you which side leans; size tells you how much.
- Antisymmetric / reflection-equivariant: swapping the input’s left and right sides flips the sign of the output. In symbols, $s(\text{swap of } x) = -s(x)$, exactly, as a fact about the readout’s wiring. The full anatomical mirror runs through the encoder first, so its measured slope is only approximately -1.
- Inductive bias: a built-in assumption that shapes what a model prefers to learn, before it sees any data.
- Probe: a small, simple rule (here a linear one) trained on top of the frozen model’s fingerprints to read out one quantity, like the signed axis.
- Transductive: the model was trained on the very videos you later test it on, so a good score can just be memorizing.
- Source-video-disjoint: no clip from a held-out video is used to train the probe, so the test video is genuinely new to the probe.
- R-squared: a score from 0 to 1 for how well a prediction matches the truth. 1 is perfect, 0 is no better than always guessing the average.

Reading the math (antisymmetric by construction). “Antisymmetric” describes a number $s(x)$ computed from an input x such that swapping the input’s two sides negates the number. - Write the input SWAP as S (exchange each left joint’s feature with its right partner’s, at the readout’s input). Antisymmetric means $s(S\ x) = -s(x)$ for every input x , exactly. - “By construction” means the equation holds because of how the readout is wired, not because a training penalty nudged it there. It is true even for inputs the model never saw. This is the exact guarantee. - The full anatomical mirror M (negate the sideways coordinate AND swap left/right landmarks on the raw skeleton) is a related but DIFFERENT operation: it passes through the encoder before the readout, so $s(M\ x)$ only approximately equals $-s(x)$, and its slope is a measured number, not exactly -1. - If $s(S\ x)$ equaled $+s(x)$ instead, the readout would be side-blind (symmetric): a swap would leave it unchanged, so it could not tell left from right. - The value $s(x) = 0$ is the only value that is its own negative (because $-0 = 0$), so a perfectly symmetric walker maps to exactly 0 on this axis, and the sign of a nonzero value says which side leans.

The claim we test is that this built-in bias is the CORRECT one for telling lateralized conditions from symmetric ones, and that building it in beats letting a standard model learn it. The negative class in this test is myopathy, which by mechanism should sit near zero on a signed axis.

Why this matters

Item 05 treats reflection-equivariance as something to measure after the fact: does the frozen `ea59fea0` model happen to flip its decoded sign under a mirror? This proposal treats reflection-equivariance as a design choice to install first, and then judge on whether it helps.

The scientific question is an inductive-bias question, and that is exactly the kind of question that carries beyond one dataset. An inductive bias is a built-in assumption that shapes what a model prefers to represent. (Locatello et al., ICML 2019, argue that pulling out clean structured factors needs such a bias or extra supervision, not just more data.) Here the assumption is a real symmetry of the world: laterality is a signed quantity, and a left-right mirror should flip its sign.

A positive result would confirm a specific, transferable belief: for models meant to separate one-sided problems from both-sides problems, wiring the signed left-minus-right axis to be antisymmetric by construction beats letting a standard model learn it, on the same frozen instrument, at a pre-registered margin. That is a claim about how to BUILD gait models, not a claim about one small group’s accuracy.

A null result rules out an equally specific belief, and it is genuinely useful. If the reflection-equivariant model does NOT beat the standard `ea59fea0` model on item 05’s signed-decodability, and does not sharpen the lateralized-versus-symmetric contrast, then the built-in constraint bought nothing here. Either the standard model already learned the symmetry well enough (remember flip is off, `flip_probability` 0.0, see Method), or the signed axis is simply not the bottleneck for this separation on $n=18$ sources. Both readings retire the easy intuition that “just add the equivariance” is a free win at this scale. ICLR/ICML/NeurIPS 2026 reviewer guidance explicitly values a well-motivated study that contributes new knowledge, including a careful negative result.

What this does NOT do: it does not turn $n=18$ sources into a claim about diagnosing anyone. The neuroscience DEFINES the target (a signed axis that separates one-sided from both-sides mechanisms) and the falsifiable prediction (equivariant beats standard on that axis). It does not license any statement about diagnosing an individual. Any clinical-accuracy statement would be external-cohort reach-tier only, and no participant-disjoint skeleton cohort for hemiplegic CP or myopathy exists to support one (see Controls and Responsible use).

Background and related work

S-JEPA is a Joint-Embedding Predictive Architecture for skeletons (Abdelfattah and Alahi, S-JEPA, ECCV 2024, DOI 10.1007/978-3-031-73411-3_21), part of the JEPA family that started with images and video (Assran et al., I-JEPA, CVPR 2023, arXiv:2301.08243; Bardes et al., V-JEPA, 2024, arXiv:2404.08471). Here are the moving parts from scratch, because the architectural change lands right inside them.

A TOKEN is the model’s smallest input unit: one BlazePose joint (Grishchenko et al., BlazePose GHUM, 2022, arXiv:2206.11678) watched over a short window of time. Each clip is stretched or squeezed to 64 frames, then 4 next-door frames form one time patch, giving 16 time positions. With 33 joints that is $33 \times 16 = 528$ possible joint-time tokens.

Reading the math (token count). This says the total number of joint-time tokens is joints times time positions. - 33 is the number of BlazePose joints. - 16 is the number of time positions (64 frames split into groups of 4). - “x” means multiply. $33 \times 16 = 528$, so there are 528 tokens. - Fewer frames per patch would give more time positions and a larger token count.

Each token turns a 4-frame by 3-coordinate (x, y, relative z) 12-number vector into a 96-number embedding through a linear layer. There are two encoders. The VIEW (online) encoder sees only the visible tokens and is trained by gradient descent (the standard way models learn, by nudging weights to lower error). The TARGET encoder sees all 528 tokens, is not trained by backpropagation, and its weights are an exponential moving average (EMA, a slowly-updated copy) of the view encoder (momentum on a cosine schedule, from 0.999 toward 1.0). A PREDICTOR (a small 2-layer Transformer with a learned mask token) guesses the target encoder’s hidden features at the masked positions. Masking-and-predicting is like covering part of a photo with your hand and guessing what is behind it. Only 12 landmarks are ever hideable as prediction targets: left/right shoulder (11,12), hip (23,24), knee (25,26), ankle (27,28), heel (29,30), foot index (31,32). The training loss (the error the model tries to shrink) is:

$$L = L_JEPA + 0.05 * L_VICReg + 0.25 * L_group$$

Reading the math (total training loss). This says the total training error is a weighted sum of three parts. - L is the total error the optimizer makes as small as it can (smaller is better). - L_JEPA is the main prediction error (weight 1, the biggest term): how badly the predictor guessed the hidden features. - L_VICReg is an anti-collapse penalty (weight 0.05). - L_group is a label-aware term (weight 0.25). - * means multiply and + means add. Both extra weights are below 1, so they are gentle nudges; 0.25 pushes five times harder than 0.05 but both stay under the main term. - Setting 0.05 to zero removes the anti-collapse guard; setting 0.25 to zero removes the label-aware pull, and Stages 1 to 4 stop being supervised fine-tuning.

VICReg (Bardes, Ponce, LeCun, ICLR 2022, arXiv:2105.04906) keeps features spread across many independent directions instead of collapsing to one boring vector. Without it, a lazy model can score well by making every fingerprint nearly identical. L_group is a label-aware term active only in Stages 1 to 4, which is why those stages count as supervised fine-tuning.

One augmentation fact is load-bearing here. The training-time geometric augmentation includes a small y-axis rotation (max 8 degrees) and small shifts, but laterality FLIP is OFF (flip_probability 0.0), precisely because left-right identity matters for stroke.

Reading the math (flip_probability 0.0). This says how often the pipeline mirrors an input left-to-right during training. - flip_probability is a chance, so it runs from 0 to 1 (0 means never, 1 means always). - 0.0 means the pipeline never flipped left and right during training. - If this were above 0, the model would be taught to treat left and right as interchangeable, erasing exactly the signed asymmetry this axis needs. We hold it at 0.0 for both encoders so the comparison is about architecture, not augmentation.

The code defines the anatomical mirror as the exact LEFT_RIGHT_PAIRS list (shoulder 11/12, hip 23/24, knee 25/26, ankle 27/28, heel 29/30, foot index 31/32, plus face and arm pairs), which negates the x coordinate and swaps each left landmark with its right partner. The architectural change and the mirror test both reuse this exact operation.

The biology grounding is the symmetry axis. LATERALIZED (one-sided) conditions raise a signed left-right difference: stroke via the crossover of motor wires (PMID 30571044, 30521239), with the validated Symmetry Ratio on step length, swing time, and stance time as the biomarker (Patterson et al. 2010, PMID 19932621); hemiplegic CP via a one-sided injury (PMID 19081519, 17261726), where the within-CP hemiplegic-versus-diplegic split IS the one-sided-versus-both-sides axis; early PD via one-sided onset (PMID 22367437), even though PD’s dominant validated biomarker is stride-time CV (Hausdorff 1998, PMID 9613733; Schaafsma 2003, stride-time CV 8.8% fallers vs 4.2% non-fallers, PMID 12809998). The NEGATIVE class is myopathy: symmetric muscle disease (Barohn 2014, PMID 25037080) with no significant left-right asymmetry versus controls (Xiong 2023, PMID 37525241) and preserved cadence and anterior

pelvic tilt (Vandekerckhove 2022, anterior pelvic tilt 16.4 vs 11.6 deg, PMID 35721358). These signed spatial, timing, and side-view joint quantities are the ones a skeleton can actually recover (Stenum 2021 temporal MAE 0.02 s/step, side-view hip/knee/ankle MAE 4.0/5.6/7.4 deg, PMID 33891585). What skeletons CANNOT recover, and this proposal does not claim, is forces or push-off (Bowden 2006), muscle-electrical activity or stiffness, twisting rotation, or a muscle-disease diagnosis.

The equivariance idea is standard in representation learning: encode a known symmetry of the world as a wiring constraint rather than hoping the model learns it. Item 05 tested reflection-equivariance as an after-the-fact property of the frozen `ea59fea0` model. This proposal makes it a design choice and measures its payoff on the same instrument. TRANSDUCTIVE means the model saw the evaluation rows during training; all readouts here are transductive, and even a held-out probe split is transductive if the model saw that video’s clips (Kapoor and Narayanan, arXiv:2207.07048; Varoquaux, NeuroImage 2018). SOURCE-VIDEO-DISJOINT means no clip from a held-out source video is used to fit the probe. Data provenance: the canonical cohort is 96 sequences from 18 source videos (normal 12 clips from 1 video, Parkinson’s 9 from 2, stroke 12 from 3, myopathic 47 from 10, cerebral palsy 16 from 2); the wider curriculum is 159 sequences from 35 source videos.

Method

This is an ambition-first, effort-HIGH item: it changes the architecture and RETRAINS the encoder, then evaluates with item 05’s frozen instrument. Everything after training reuses the item 05 code.

The architectural lift is a reflection-equivariant readout head over paired joints. Instead of dumping all tokens into one vector and hoping a linear probe finds a signed axis, we do this: for each `LEFT_RIGHT_PAIRS` entry, pass the left joint and the right joint through the SAME shared per-joint map, take the DIFFERENCE between the two results, and add up those differences into the signed axis. Because the two sides go through the identical map and only their difference survives, mirroring the input (which swaps left and right and negates x) flips the axis exactly.

Reading the math (the antisymmetric head). Let f be a single shared function applied to each hideable joint’s token feature, and let L_k, R_k be the left and right joint features of pair k . The signed axis is $s = \text{sum over } k \text{ of } (f(L_k) - f(R_k))$. - f is shared: the same f acts on the left joint and on the right joint, so the two sides are treated identically before the subtraction. - Only the DIFFERENCE enters s . There is deliberately no $f(L_k) + f(R_k)$ term: a `left + right` sum does not change when you swap left and right, so it is mirror-invariant and would dilute the guaranteed sign-flip. The head keeps the difference and drops the sum. - Write S for the SWAP that exchanges L_k with R_k at the head’s input. After the swap the term becomes $f(R_k) - f(L_k)$, the negative of the original, so $s(S\ x) = -s(x)$ EXACTLY, for every input, trained or not. This is the by-construction guarantee, and it is a fact about the head’s algebra alone. - The full anatomical mirror M (negate the sideways coordinate AND swap left/right landmarks on the RAW skeleton) is different: it runs back through the encoder before reaching the head, and the encoder mixes tokens and adds per-joint position codes, so $s(M\ x)$ is only APPROXIMATELY $-s(x)$. Its slope is a measured number (-0.741 for the standard `ea59fea0` model), never exactly -1. Keep the two apart: the swap S is the exact wiring guarantee; the anatomical mirror M is the measured behavior of the whole model. - A perfectly symmetric walker gives $f(L_k) = f(R_k)$ for each pair, so every term is 0 and $s = 0$. Myopathy should sit near 0; lateralized conditions should sit away from 0 with a consistent sign.

We keep the S-JEPA backbone unchanged (one small Transformer, 4 heads, GELU, pre-norm; its width and depth are always read from the checkpoint `config`, not hardcoded, and the real `ea59fea0` checkpoint is `embed_dim 96`, encoder depth 4, predictor depth 2) and add the antisymmetric head as the readout used to define and score the signed axis. We RETRAIN the full five-stage curriculum matching the `ea59fea0` lineage: Stage 0 normal 300 epochs, then Stages 1 to 4 add PD, stroke, myopathic, cerebral palsy at 75 epochs each, for 600 curriculum epochs and 11,400 optimizer updates, on the 96 canonical (up to 159 curriculum) sequences, single-GPU feasible. `flip_probability` stays 0.0 throughout, for BOTH the new equivariant encoder and the standard `ea59fea0` baseline, so the comparison isolates the architectural constraint, and the no-flip rule protects lateralized asymmetry (that protection is the whole point).

Pseudo-code for the antisymmetric head and the by-construction mirror guarantee:

```
import torch

# LEFT_RIGHT_PAIRS: (left_joint_index, right_joint_index), the exact anatomy.
LEFT_RIGHT_PAIRS = [(11, 12), (23, 24), (25, 26),
                    (27, 28), (29, 30), (31, 32)]

def signed_axis(token_features, f):
```

```

# token_features: dict joint_index -> feature vector from the encoder.
# f: a SHARED per-joint map (the same f for left and right).
# Only the DIFFERENCE enters s; there is no (f(L) + f(R)) term, because a
# left+right sum is mirror-invariant and would break the sign-flip.
s = 0.0
for left_idx, right_idx in LEFT_RIGHT_PAIRS:
    s = s + (f(token_features[left_idx]) - f(token_features[right_idx]))
return s          # antisymmetric by construction (under the input swap)

# WIRING check (exact): swap left/right features at the head's input, so the
# axis must negate. signed_axis(swap(features)) == -signed_axis(features)
# holds to floating-point precision, because each (f(L) - f(R)) term becomes
# (f(R) - f(L)). This is algebra about the head, independent of the encoder.
#
# ANATOMICAL-mirror check (measured, NOT exact): mirror the RAW skeleton and
# re-run it through the encoder before the head. The encoder mixes tokens and
# adds per-joint position codes, so the mirrored features are not a clean swap
# of the originals; the resulting slope is a measured number (-0.741 for
# the standard ea59fea0 model), never exactly -1.

```

Evaluation reuses item 05's frozen instrument without change. Cache the 528-token target-encoder features for each sequence from the RETRAINED equivariant encoder and, separately, from the standard `ea59fea0` encoder. Fit item 05's ridge probe for signed-decodability (a linear rule from features to the signed target, with a small penalty that keeps its weights modest, chosen on training sources only), and report held-out-source R-squared and mean absolute error (the average size of the miss, in target units), against item 05's two reference bounds: the RAW-COORDINATE NULL (the same fit on handcrafted signed left-minus-right coordinate features, no network at all) and the UNTRAINED-ENCODER FLOOR (the same probe on a random, untrained encoder of identical shape). Then run item 05's mirror-slope test: apply the exact `LEFT_RIGHT_PAIRS` mirror, re-embed, decode, and fit the slope of decoded-mirrored versus decoded-original against the ideal line $y = -x$.

Reading the math (the mirror line $y = -x$ and its slope). The ideal mirror response is: mirrored output equals the negative of the original output. $-x$ is the decoded number on the original input; y is the decoded number on the mirrored input. - A perfect flip has slope -1: same size, opposite sign, so all points fall on the line $y = -x$. - The slope is exactly -1 only for the WIRING check, where left and right are swapped at the head's input (algebra about the head alone). When the ANATOMICAL mirror is applied to the raw skeleton and re-run through the encoder, the slope is a MEASURED, approximate quantity for every model, including the equivariant one, because the encoder mixes tokens and adds position codes. Item 05 measured -0.741 for the standard `ea59fea0` encoder, and Arm 1's antisymmetric head measured -0.223 on the same encoder. We report the measured anatomical slope and never call it -1.

The decisive experiment

The split is stated before any fitting. Folds are SOURCE-VIDEO-DISJOINT: signed laterality is pooled across all conditions and whole source videos are held out, never clips. Because per-condition source counts are tiny (normal 1 source, Parkinson's 2, stroke 3, myopathic 10, cerebral palsy 2), we do NOT report per-class leave-one-source-out R-squared on $n=1$ held-out sources; the primary endpoint is pooled across conditions, every source video is its own dot, and a source-level permutation null is used only where the number of held-out sources makes it meaningful. The primary comparison runs on a PROVENANCE-MATCHED subset (all canonical-path sequences), because most normal rows use the augmented extraction path while every abnormal row uses the canonical path, so a naive contrast could learn how the videos were processed rather than how people walked.

Primary endpoint: held-out-source signed-decodability R-squared of the RETRAINED reflection-equivariant encoder MINUS that of the standard `ea59fea0` encoder, both read with item 05's identical frozen probe and both reference bounds.

Pre-registered margin: the reflection-equivariant encoder must exceed the standard `ea59fea0` encoder by at least 0.05 held-out-source R-squared on the signed axis AND must itself clear item 05's own bar (above the untrained-encoder floor by at least 0.05 R-squared and at least 80 percent of the raw-coordinate null), with the sign of the decoded number consistent on at least 75 percent of held-out sources. Falling short is scored as an informative null: the architectural constraint did not beat the learned behavior at this scale.

Note: this proposal-level margin was a first draft, and it is SUPERSEDED. Do not apply the 0.05-over-

standard, 80-percent-of-null, and 75-percent-sign gates to any real number; they were replaced before any real run. The REIFIED code enforces a hardened rule instead: the binding bar is $\max(D, C)$ (the untrained floor C is the real constraint, because in item 05 C at -0.156 already beat the learned encoder D at -0.602 by 0.446); the fixed 75-percent sign threshold is replaced by a source-label permutation null on the head; a capacity-matched control Ac (the same head with the symmetric left-plus-right path added) must be beaten by 0.05 so the win is attributable to the antisymmetry constraint rather than to head capacity; the nuisance control is an absolute $\text{abs}(E) < 0.05$ on a Lane E symmetrized over the anatomical mirror so it is genuinely side-blind; and the unreachable “80 percent of the raw-coordinate null” gate is dropped. The floor C ’s own permutation null is still reported, but only as floor characterization (a strong floor just raises the binding bar), not as a claim-withdrawing gate. See IMPLEMENTATION.md section 5 for the authoritative rule.

Reading the math (the margin numbers). A positive result needs all thresholds at once. - 0.05 R-squared is the smallest advantage of the equivariant encoder over the standard one that counts as a real gain; below it the architecture bought nothing measurable. - 0.05 R-squared over the untrained floor is the smallest gap that counts as beating chance for the equivariant encoder on its own. - 80 percent (a fraction of 0.80) is the share of the raw-coordinate ceiling the equivariant probe must reach to count as competitive with the non-neural baseline. - 75 percent (a fraction of 0.75) is the share of held-out sources whose decoded sign must point the correct way. - If any threshold is missed, the run is an informative null.

Worked example (illustrative numbers only, not measured facts). Suppose on 4 held-out source videos the standard `ea59fea0` probe scores R-squared 0.36, the equivariant encoder scores 0.44, the untrained floor scores 0.05, and the raw-coordinate null scores 0.50. - Beat the standard encoder by at least 0.05: 0.44 minus 0.36 = 0.08, above 0.05. Pass. - Beat the floor by at least 0.05: 0.44 minus 0.05 = 0.39, far above 0.05. Pass. - Reach at least 80 percent of the null: 0.80 times 0.50 = 0.40, and 0.44 is above 0.40. Pass. - Sign consistency: decoded sign matched on 3 of 4 sources, 3 / 4 = 0.75, meets the bar. Pass. - All pass, so this illustrative run would support “reflection-equivariance is the correct inductive bias, and building it in beats learning it.” If instead the equivariant encoder scored 0.39, then 0.39 minus 0.36 = 0.03 is below 0.05, and the run would be scored an informative null even though 0.39 clears the floor.

Lateralized-versus-symmetric secondary endpoint: on the provenance-matched canonical subset, the signed axis on lateralized-labelled sources (stroke, hemiplegic-labelled CP, PD) should sit away from zero with a consistent sign, and on symmetric-labelled sources (myopathic) should sit near zero, matching the mechanism (Xiong 2023, PMID 37525241; Patterson 2010, PMID 19932621). This is reported as a mechanism check with every source as a dot, never as a per-individual diagnosis.

Simple non-neural / nuisance baseline: the raw-coordinate null (Lane B) is the non-neural baseline. The pooled control (Lane E) is the nuisance baseline. Raw mean-and-standard-deviation pooling of tokens is order-blind but NOT side-blind on its own, because the encoder’s learned per-joint position codes keep each token conditioned on its landmark slot; the reified code therefore SYMMETRIZES Lane E over the anatomical mirror, $E(x) = 0.5 * (\text{pool}(\text{encoder}(x)) + \text{pool}(\text{encoder}(Mx)))$, so $E(x) == E(Mx)$ by construction and the control is genuinely side-blind. It must NOT recover a signed axis; if it does, the “signed” claim is an artifact.

Lane	Feature source	Retrain?	Role	Expected on signed axis
A Equivariant probe	RETRAINED antisymmetric-head encoder	Yes	Primary	Beat Lane D and Ac by ≥ 0.05 R-squared; clear the untrained floor by 0.05; wiring-swap slope -1 exactly, anatomical-mirror slope measured
Ac Capacity-matched control	Same head, adds the symmetric left-plus-right path	No	Attribution control	Lane A must beat it by ≥ 0.05 (isolates the antisymmetry constraint)
B Raw-coordinate null	Handcrafted signed left-minus-right coords	No	Non-neural ceiling	Reference target

Lane	Feature source	Retrain?	Role	Expected on signed axis
C Untrained-encoder floor	Random-init encoder of identical architecture	No	Floor	Near chance; its permutation is reported as floor characterization only
D Standard encoder	Frozen ea59fea0 per-token features	No	Learned-behavior comparator	Item 05’s learned baseline
E Pooled control, mirror-symmetrized	Pooled tokens averaged over a sample and its anatomical mirror	No	Side-blind nuisance	$E(x) == E(Mx)$ exactly; must NOT recover a signed axis

Controls

- Hold `flip_probability` at 0.0 for BOTH the equivariant encoder and the standard **ea59fea0** comparator, so the only difference is the architectural constraint, not the augmentation. Any advantage cannot be an “the equivariant model saw mirrored data” artifact.
- Bind every number to ONE fingerprint before comparing. The standard baseline is the **ea59fea0** lineage; the equivariant encoder gets its own recorded fingerprint. Do not mix the canonical **dba24a** lineage or the superseded **d0acc262** lineage into the same comparison.
- Provenance-matched primary comparison on the canonical-path subset, since normal rows are mostly augmented-path and abnormal rows are canonical-path; a naive normal-versus-abnormal contrast could learn how videos were processed (embedding-level normal-versus-abnormal separability is around 0.96 AUC, the number the provenance confound puts at risk).
- Mirror-symmetrized pooled negative control (Lane E) that must NOT recover a signed axis. Raw pooling throws away token order but, because of the encoder’s per-joint position codes, not side identity; averaging the pooled feature over a sample and its anatomical mirror makes $E(x) == E(Mx)$ exactly, so the control is genuinely blind to a body-versus-reflection flip. The code asserts this invariance before trusting the gate.
- Untrained-encoder floor (Lane C) and raw-coordinate null (Lane B) reused unchanged from item 05, so the equivariant encoder is credited only when it clears the same bars item 05 set.
- No per-class LOSO R-squared margins on $n=1$ held-out sources; signed laterality is pooled across conditions, every source is a dot, source-level permutation only where meaningful.
- Transductive caveat next to every number: all readouts are transductive, and a held-out probe split is still transductive if the encoder saw that video’s clips. Seed variation is not source variation.
- Equivariance manipulation check: verify numerically that the equivariant head’s WIRING-swap slope is -1 to floating-point tolerance (left and right features swapped at the head’s input), confirming the by-construction guarantee before any comparison is trusted. This is separate from the measured anatomical-mirror slope, which is not expected to be -1.
- Honest external-cohort limitation: no participant-disjoint skeleton cohort for hemiplegic CP or myopathy exists, so the lateralized-versus-symmetric separation stays transductive and internal. The only bounded external anchors are method-level: CASIA-B (Yu 2006), OU-MVLP-Pose (Takemura 2018), GREW (arXiv:2205.02692), Gait3D (arXiv:2204.02569) can test whether the built-in mirror-equivariance property holds across camera viewpoints and mirrored views (an architecture check, non-clinical); Human3.6M (Ionescu 2014, DOI 10.1109/tpami.2013.248) against motion capture checks that the pose landmarks feeding the signed axis are geometrically faithful. The PD stride-time-variability biomarker has a label-level cross-modal anchor in PhysioNet Gait-in-PD (gaitpdb, 93 PD + 73 controls, Hausdorff, DOI 10.13026/C24H3N), but that is force/IMU, not skeleton, and speaks to the variability axis, not this signed axis.

How this differs from the existing plan

The nearest neighbor is item 05, which this proposal deliberately uses as its EVALUATION INSTRUMENT rather than duplicating. Item 05 asks whether the standard frozen **ea59fea0** encoder HAPPENED to learn an antisymmetric signed axis (a property measured after the fact). This proposal BUILDS reflection-equivariance into the architecture (antisymmetric by construction, so the wiring-swap slope is -1 exactly while the anatomical-mirror slope stays a measured number) and RETRAINS, then asks whether that built-in bias BEATS the learned behavior on item 05’s own frozen probe and bounds. Item 05 does not retrain and does not compare a constrained encoder against the standard one; this

item does both.

Against the wider portfolio: plan/04 retrains encoders to ablate the prediction TARGET (motion versus position); this item retrains to change the READOUT symmetry, not the target. Plan/05 varies the pooling operator and defines signed ankle phase-lag targets; this item keeps item 05's single clean signed-laterality target and does not re-derive phase-lag. Plan/07 stresses viewpoint invariance broadly; this item installs one specific equivariance (the anatomical left-right mirror) as a mechanism-aligned inductive bias, not a general robustness sweep. Item 06 treats provenance as a nuisance control; here provenance-matching is a control, not the object. No existing plan or ideas item builds reflection-equivariance into the encoder and tests whether the architectural constraint beats the learned one.

Timeline (feasibility-tiered, ambition-first)

Core tier (effort HIGH, may exceed 3 weeks because it retrains the curriculum).

Week 1 (16 to 22 Aug 2026): implement the antisymmetric head over LEFT_RIGHT_PAIRS, prove the by-construction WIRING guarantee numerically (the head's slope is exactly -1 to tolerance when left and right features are swapped at its input), wire it into the S-JEPA backbone (shape read from the checkpoint config, embed_dim 96, encoder depth 4 for the real lineage), and confirm the canonical parquet carries source, condition, and provenance columns before any join. Freeze item 05's signed-laterality target function and source-video-disjoint fold manifest unchanged.

Day-5 gate (20 Aug 2026): continue only if the equivariant head's mirror slope is -1 to floating-point tolerance (the guarantee holds), the standard ea59fea0 baseline cache is bound to a single fingerprint, the provenance-matched canonical subset is assembled, and no held-out source's clips leaked into the fold being read.

Week 2 to Week 3 (23 Aug to 5 Sep 2026): RETRAIN the five-stage curriculum for the equivariant encoder (Stage 0 normal 300 epochs, Stages 1 to 4 add PD, stroke, myopathic, cerebral palsy 75 epochs each, 600 epochs, 11,400 updates, flip_probability 0.0), monitor representation health (feature std, effective rank, mean pairwise cosine) against the ea59fea0 reference figures, and cache the 528-token features. Fit Lanes A, B, C, D, E on source holdouts; run the mirror-slope check on all lanes; assemble per-source dots and the source-level permutation null where meaningful.

Day-14 gate (29 Aug 2026): continue to write-up only if the equivariant encoder trained without collapse (feature std well above zero, non-degenerate effective rank), the primary equivariant-minus-standard endpoint has a clean verdict (clearing the pre-registered margin or an interpretable null), and Lane E correctly fails to recover a signed axis.

Week 4 (6 to 12 Sep 2026): produce the figures, finalize the equivariant-versus-standard signed-decodability table, the lateralized-versus-symmetric source-level panel, and the mirror-slope manipulation check, write transductive caveats next to every number, and package the head implementation, retrained fingerprint, fold manifest, and per-source results.

Reach tier (+2 to 3 weeks, marked honestly): the multi-view mirror-consistency check on CASIA-B and OU-MVLP-Pose as a method-level external anchor (does the built-in equivariance hold across camera viewpoints and mirrored views on non-clinical multi-view pose). No new clinical data is used or exists; reach anchors use existing public non-clinical multi-view pose only. All results remain transductive; source video is the independent unit; folder labels are dataset annotations, not diagnoses.

Figures

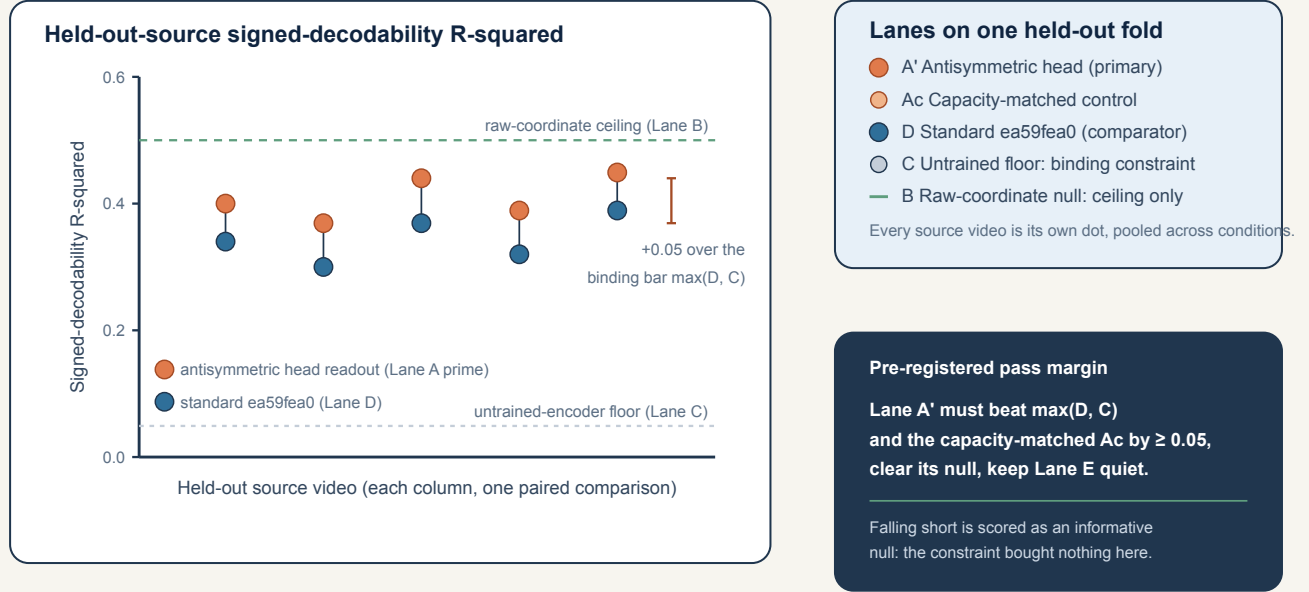
Fig 1: per held-out source, the signed-decodability R-squared of the by-construction reflection-equivariant encoder against the standard ea59fea0 encoder on item 05's frozen instrument, paired dot by dot, with the pre-registered 0.05 R-squared advantage drawn as the pass band. The by-construction encoder must beat ea59fea0 by at least that margin. How to read this picture: each pair of dots is one held-out video, one for the new encoder and one for the old; higher is better, and the shaded band shows the 0.05 head start the new encoder has to win by.

Fig 2: the mirror-slope check, which has two distinct panels. The WIRING check swaps left and right features at the head's input and lands exactly on $y = -x$ (slope -1 to floating-point tolerance), because that is algebra about the head alone. The ANATOMICAL-mirror check mirrors the raw skeleton and re-runs it through the encoder, giving a MEASURED slope that is only approximately -1 for every model (-0.741 for the standard ea59fea0 encoder). The exact line belongs to the wiring check; the anatomical slope is reported honestly as measured. How to read this picture: if the model behaved like a perfect left-minus-right scale, mirroring the person would flip the reading and the dots would fall on the diagonal from top-left to bottom-right; the wiring check sits exactly on it by construction, while the anatomical-mirror dots sit only as close as the trained encoder happens to be.

Fig 3 (beginner concept diagram): a friendly picture of the whole idea. On the left, a walker with a stronger left side gives a positive reading on the signed left-minus-right scale. Flip that walker in a mirror and the same honest scale reads

Antisymmetric head (A prime) vs the binding bar max(D, C)

Illustrative expected shape. Paired dots, one held-out source per column, signed R-squared on item 05's frozen probe. Arm 1 reads ea59fea0 as is; Arm 2 retrain. Transductive.



Illustrative expected shape, not measured results. Provenance-matched canonical subset. Item 05 asymmetry readout R-squared ≈ 0.154 . Labels are dataset annotations.

Figure 1: Signed-laterality decodability: retrained reflection-equivariant encoder vs standard ea59fea0, per held-out source, with the pre-registered advantage annotated

the exact negative. A symmetric walker sits at zero. How to read this picture: follow the arrow through the mirror and watch the sign of the number flip from plus to minus; that sign-flip is what “reflection-equivariant” means, and a both-sides-equal walker parks on the zero mark in the middle.

Fig 4 (beginner concept diagram): the difference between building the rule in and hoping it is learned. One path shows the antisymmetric head, where left and right go through the same box and only their difference comes out, so a mirror must flip the sign every time. The other path shows a standard encoder that is only nudged toward that behavior by training. How to read this picture: the top path is a guarantee (it works for every input, even ones never seen), the bottom path is a hope (it works only as far as the training taught it); this proposal tests whether the guarantee beats the hope.

Responsible use

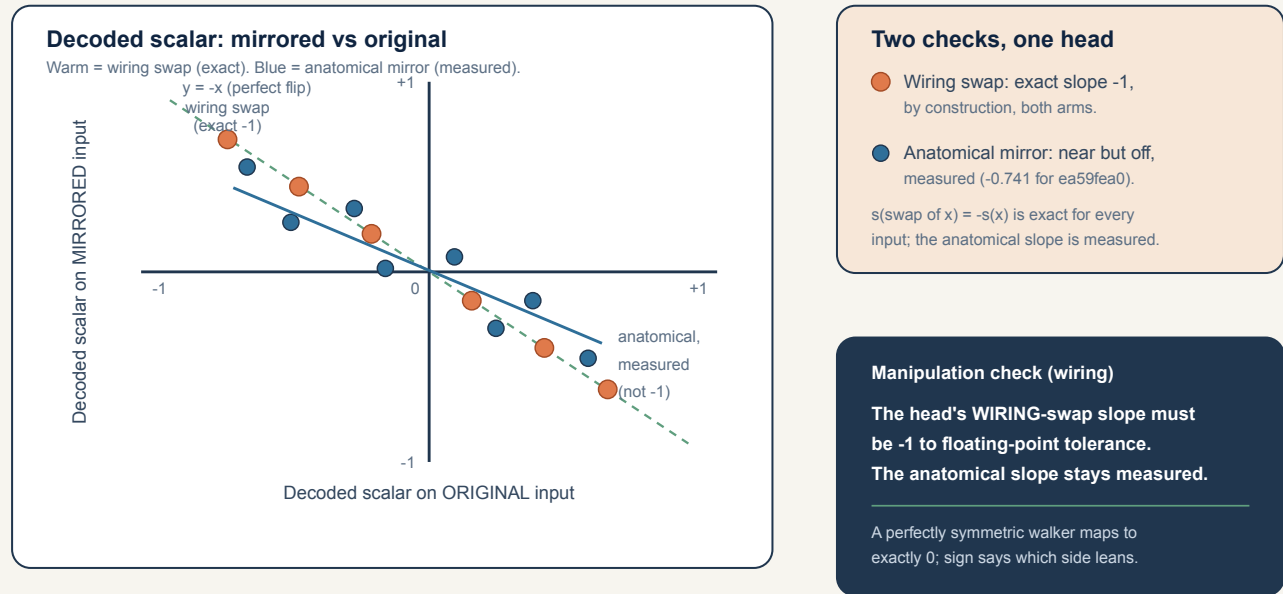
The condition folder labels (normal, parkinsons, stroke, myopathic, cerebralpalsy) are dataset annotations from GAVD (Ranjan et al., IEEE Access 2025, DOI 10.1109/ACCESS.2025.3545787), not diagnoses made by this project. Terms like “lateralized” and “symmetric” here describe a skeleton-measurable signed axis grounded in the cited mechanisms, not a clinical judgment about any individual. The signed axis is a representation diagnostic computed from cached skeleton coordinates; it is not a validated clinical biomarker and must not be read as a measurement of any person’s health. All results are transductive and small-sample (18 canonical source videos), with the source video as the independent unit. The neuroscience defines the target and the falsifiable prediction; it does not upgrade $n=18$ sources into a clinical-accuracy claim. Any clinical-accuracy statement would be external-cohort reach-tier only, and no participant-disjoint skeleton cohort for hemiplegic CP or myopathy exists to support one. Skeletons cannot recover forces or push-off, muscle-electrical activity or stiffness, twisting rotation, or a muscle-disease diagnosis, and this proposal claims none of these.

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Two mirror checks against the $y = -x$ reflection line

Illustrative expected shape. Exact wiring swap (slope -1 by construction) vs measured anatomical mirror through the encoder. Transductive.

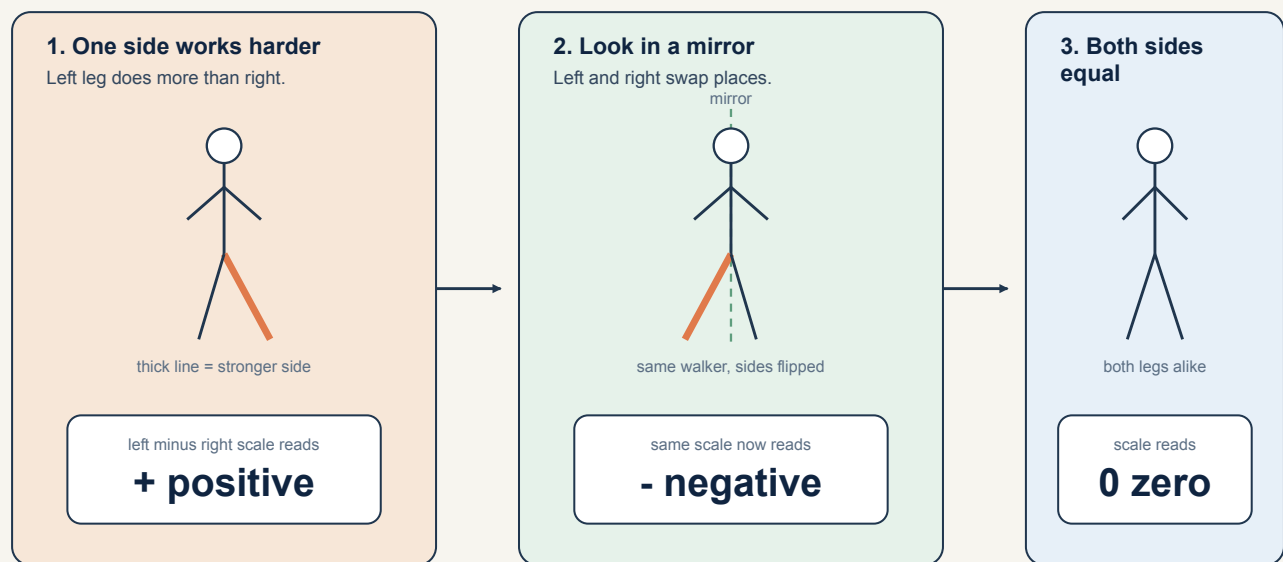


Illustrative expected shape, not measured results. A mirror flips the sign of the lateralized biomarker (Patterson 2010). Labels are dataset annotations.

Figure 2: Mirror-slope check: decoded scalar on original versus anatomically mirrored input against the $y = -x$ reflection line

A mirror flips the sign of a left-minus-right scale

Read left to right. An honest "left minus right" scale must flip its reading when you flip the walker in a mirror.

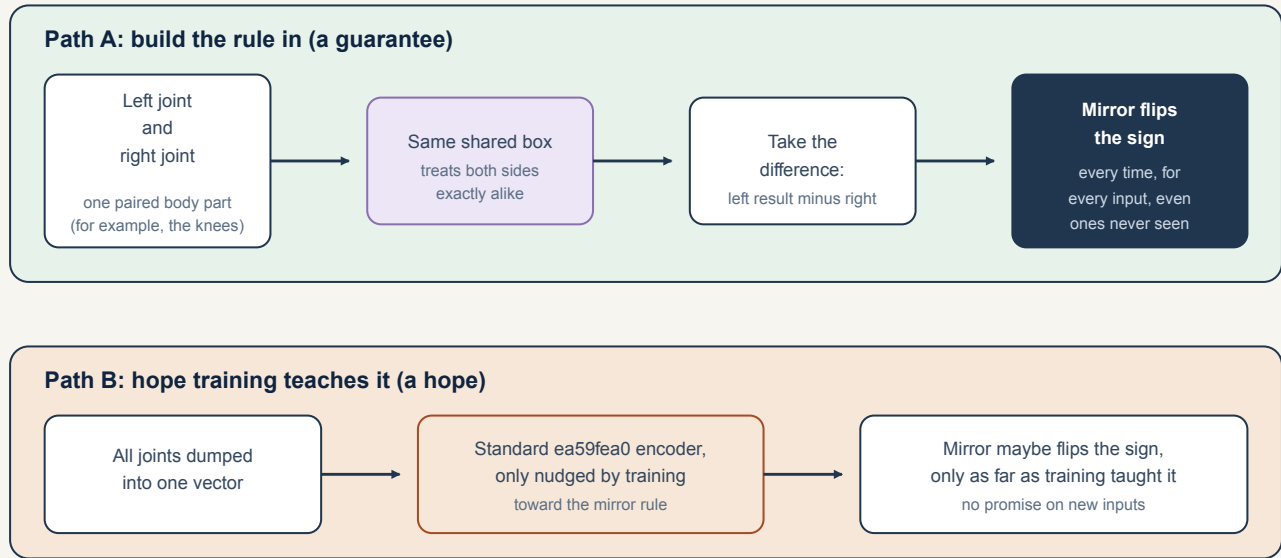


That guaranteed flip (plus to minus, same size) is what "reflection-equivariant" means. A both-sides-equal walker parks at exactly zero.
Folder labels (normal, parkinsons, stroke, myopathic, cerebral_palsy) are dataset annotations from GAVD, not diagnoses made here.

Figure 3: Beginner concept diagram: a mirrored walker flips a signed left-minus-right scale from positive to negative

Built-in guarantee vs hoping the model learns it

Two ways to get the mirror rule. Top path forces it by wiring; bottom path only nudges toward it in training.



Both encoders keep left-right flip OFF in training (flip probability 0.0), so the only difference is the wiring. This proposal tests whether the guarantee beats the hope. Folder labels (normal, parkinsons, stroke, myopathic, cerebral_palsy) are dataset annotations from GAVD, not diagnoses made here.

Figure 4: Beginner concept diagram: built-in wiring versus hoping the model learns the mirror behavior

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